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BALLARI INSTITUTE OF TECHNOLOGY & MANAGEMENT

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(Recognized by Govt. of Karnataka, approved by AICTE, New Delhi & Affiliated to Visvesvaraya Technological University, Belagavi)
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DEPARTMENT OF CSE(DATA SCIENCE)

A Mini-Project Report On

"REAL ESTATE DATA ANALYSIS"

A report submitted in partial fulfillment of the requirements for the

DATA VISUALIZATION WITH POWER BI

Submitted By

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Belagavi, Karnataka 2025-2026

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CERTIFICATE

This is to certify that the OPEN-ENDED PROJECT of DATA VISUALIZATION WITH POWER BI **"REAL ESTATE DATA ANALYSIS"** has been successfully presented by **K.DEEPIKA** bearing USN **3BR23CD036** student of semester B.E for the partial fulfillment of the requirements for the award of **Bachelor Degree in CSE(DS)** of the BALLARI INSTITUTE OF TECHNOLOGY& MANAGEMENT, BALLARI during the academic year 2025-2026.

Signature of guide	Signature of HOD
Mr. Sundeep Sindhanoor	**Dr. Aradhana D**

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ABSTRACT

The *Real Estate Data Analysis Dashboard* project focuses on analyzing and visualizing real estate data using Power BI to generate meaningful insights for better decision-making. In today's data-driven world, the real estate industry produces large volumes of data related to property prices, customer preferences, builder performance, and transaction details. However, raw data alone is difficult to interpret and does not effectively support strategic decisions.

This project aims to transform raw data into an interactive and user-friendly dashboard that highlights key performance indicators such as total revenue, average property price, number of properties, and customer distribution. The dataset used in this project includes information about builders, customers, properties, and transactions, which are processed and analyzed using Power BI.

Data preparation is carried out using Power Query, where data is cleaned, transformed, and structured for analysis. DAX (Data Analysis Expressions) is used to create calculated measures and perform advanced computations. Various visualizations such as bar charts, line charts, pie charts, and tables are used to represent the data in a clear and understandable manner.

The dashboard also includes filters and slicers that allow users to interact with the data dynamically by selecting different areas, property types, and cities. This interactivity enhances the user experience and enables customized analysis.

The results of the project reveal important insights such as variations in property prices across different locations, revenue contribution by different builders, and trends in property sales over time. These insights can help stakeholders such as buyers, investors, and developers make informed decisions.

In conclusion, this project demonstrates the effectiveness of data visualization tools like Power BI in simplifying complex datasets and providing valuable business insights in the real estate domain.

CHAPTER 1

INTRODUCTION

1.1 About the Project

The *Real Estate Data Analysis Dashboard* project is developed to analyze and visualize real estate data in an interactive and meaningful way using Power BI. The real estate industry generates large volumes of data related to property prices, customer preferences, builder performance, and sales transactions. However, raw data alone is difficult to interpret and does not provide actionable insights.

This project focuses on converting raw real estate data into a structured and visually appealing dashboard that helps users understand trends, patterns, and key performance indicators. The dashboard presents important metrics such as total revenue, average property price, number of properties, and customer distribution.

By using data visualization techniques, the project enables users to make informed decisions regarding property investments, pricing strategies, and market analysis.

1.2 Language Used

The project primarily uses the following tools and languages:

- **Power BI Desktop** – For data visualization and dashboard creation
- **Power Query (M Language)** – For data cleaning and transformation
- **DAX (Data Analysis Expressions)** – For creating calculated measures and KPIs
- **Microsoft Excel / Dataset** – For storing and managing raw data

These tools together help in transforming raw data into meaningful insights efficiently.

CHAPTER 2

SCOPE OF THE PROJECT

2.1 Problem Statement

In the real estate sector, data is often scattered across multiple sources and formats, making it difficult to analyze and interpret. Stakeholders such as buyers, sellers, and developers face challenges in understanding pricing trends, identifying profitable areas, and evaluating builder performance.

Traditional methods of analysis are time-consuming and lack interactivity. Therefore, there is a need for a system that can:

- Consolidate data into a single platform
- Provide real-time insights
- Allow dynamic filtering and comparison

2.2 Objectives

The main objectives of this project are:

- To analyze real estate data efficiently using Power BI
- To visualize key metrics such as revenue, pricing, and sales trends
- To compare property prices across different areas
- To evaluate builder performance based on revenue and pricing
- To provide an interactive dashboard for better decision-making

CHAPTER 3

SYSTEM REQUIREMENTS

3.1 Functional Requirements

- Load and process real estate datasets
- Perform data cleaning and transformation
- Create relationships between tables
- Generate calculated fields using DAX
- Build interactive dashboards with charts and KPIs
- Enable filtering using slicers

3.2 Non-Functional Requirements

- The system should be user-friendly
- The dashboard should load quickly
- Data should be accurate and consistent
- Visualizations should be clear and easy to interpret

3.3 Hardware Requirements

- Minimum 4GB RAM (8GB recommended)
- Processor: Intel i3 or higher
- Storage: At least 10GB free space

3.4 Software Requirements

- Power BI Desktop
- Microsoft Excel
- Windows Operating System

CHAPTER 4

METHODOLOGY

The methodology followed in this project involves a step-by-step process to collect, process, analyze, and visualize stock data effectively.

4.1 Data Collection :

The dataset used in this project includes information about properties, builders, customers, and transactions. The data was collected from structured sources such as Exc el files.

4.2 Data Preparation:

Data preparation is done using Power Query where:

- Missing values are handled
- Columns are cleaned and formatted
- Data types are corrected
- Unnecessary columns are removed

4.3 Data Visualization :

Different types of charts are used:

- Bar charts for comparisons
- Line charts for trends
- Pie charts for distribution
- Tables for detailed analysis

4.4 Filters and Interactivity:

Slicers are added for:

- City
- Area
- Property Type

These allow users to interact with the dashboard dynamically.

4.5 Dashboard Design :

The dashboard is designed with: Dashboard Design

- Clear layout
- Proper alignment of visuals
- Highlighted KPIs
- User-friendly interface

4.6 Outcome :

The final outcome is an interactive dashboard that provides:

- Clear insights into real estate trends
- Easy comparison of data
- Better decision-making support

CHAPTER 5

IMPLEMENTATION

The processed data is then visualized using Microsoft Power BI, enabling the development of an interactive and user-friendly dashboard. Various charts, including pie charts for property type distribution, bar charts for area-wise price comparison, column charts for builder performance analysis, and KPI indicators, are designed to provide clear insights into real estate trends. Line charts are used to represent sales trends over time, helping users understand market growth patterns. Filters and slicers are implemented to allow users to explore data based on different areas, property types, and cities. This interactive approach enhances data analysis and supports better decision-making for stakeholders.

5.1 Dataset Description

The dataset consists of:

- Builders table
- Customers table
- Properties table
- Transactions table

5.2 Data Loading

Data is imported into Power BI using the “Get Data” option from Excel files.

5.3 Data Transformation (Power Query)

Data is transformed by:

- Removing duplicates
- Renaming columns
- Filtering unnecessary data

\

5.4 Data Modelling

Relationships are created between tables using keys such as:

- Property ID
- Customer ID

5.5 DAX Measures (Key Calculations)

Important calculations include:

- Total Revenue
- Average Price
- Count of Properties
- Count of Customers

5.6 Dashboard Visualizations

The dashboard includes:

- KPI cards
- Bar charts
- Line charts
- Pie charts
- Tables

5.7 Filters / Slicers

Slicers allow users to filter data by:

- Area
- Property Type
- City

5.8 Dashboard Layout

The layout is structured to ensure:

- Easy navigation
- Logical grouping of visuals
- Clean design

5.9 Integration

All visuals are connected and update dynamically based on user interaction.

5.10 Final Output

The final output is a fully functional Power BI dashboard that provides meaningful insights into real estate data.

CHAPTER 6

RESULT

The developed *Real Estate Data Analysis Dashboard* successfully provides an interactive and visually rich analysis of property data across different areas and builders. The dashboard transforms raw real estate data into meaningful insights using various visualizations such as pie charts, bar graphs, line charts, tables, and KPI cards.

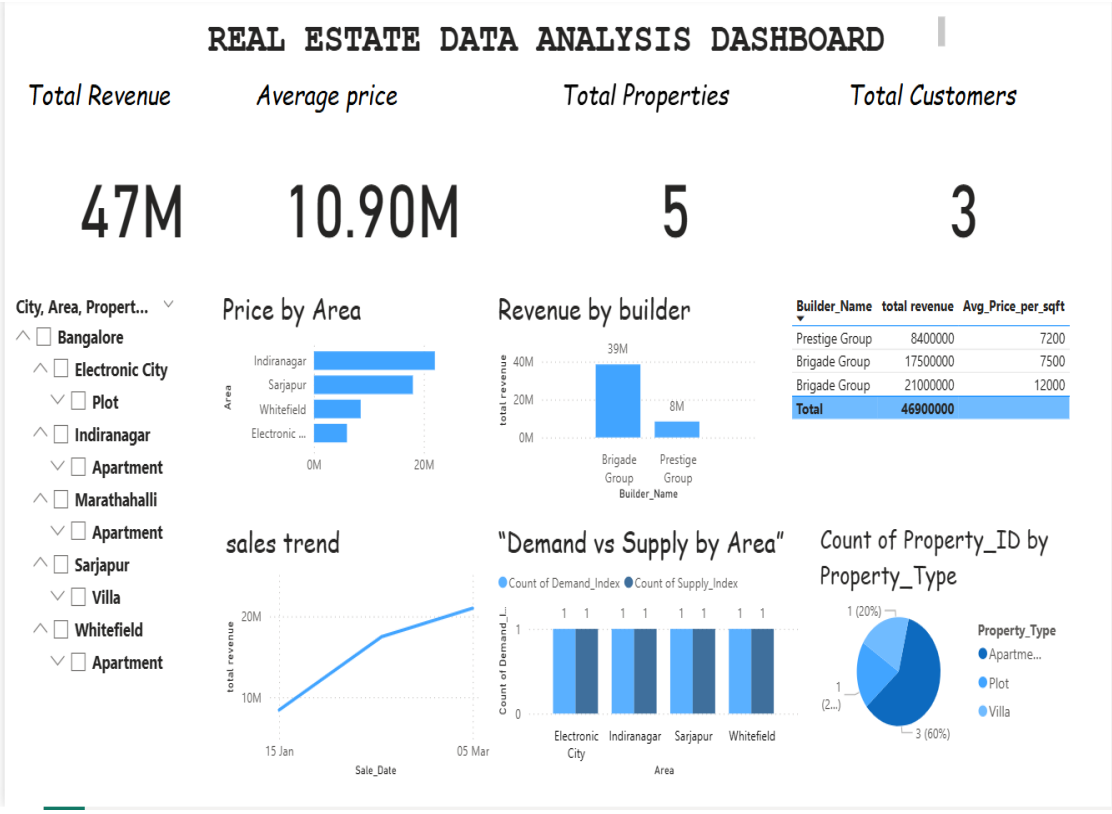
The visualizations clearly highlight variations in property prices, revenue generation, and customer distribution. Bar charts effectively represent area-wise price comparisons, while builder performance charts show revenue contributions by different developers. Line charts illustrate sales trends over time, helping users understand market growth patterns. Pie charts display the distribution of different property types such as apartments, villas, and plots.

KPI cards present key metrics such as total revenue, average property price, total number of properties, and total customers, providing a quick overview of overall market performance. These indicators help users quickly assess the status of the real estate market.

Interactive filters and slicers such as area, city, and property type allow users to customize their analysis based on specific requirements. This enhances usability and enables detailed exploration of real estate trends across different locations.

The dashboard also incorporates calculated measures such as total revenue, average price, and property count, which help in understanding pricing patterns and builder performance. The integration of multiple visuals ensures that users can analyze both individual components and overall market behavior effectively.

Overall, the project successfully achieves its objective of providing a simple, interactive, and informative platform for real estate data analysis. It helps users identify trends, compare property values, evaluate builder performance, and gain valuable insights, supporting better decision-making in the real estate domain.


REAL ESTATE DATA ANALYSIS DASHBOARD
Total Revenue
47M
Average price
10.90M
Total Properties
5
Total Customers
3
City, Area, Propert...
Bangalore
Electronic City
Plot
Indiranagar
Apartment
Marathahalli
Apartment
Sarjapur
Villa
Whitefield
Apartment
Price by Area
Indiranagar
Sarjapur
Whitefield
Electronic ...
Area
0M
20M
Revenue by builder
39M
8M
40M
20M
0M
total revenue
Brigade Group
Prestige Group
Builder_Name
Builder_Name total revenue Avg_Price_per_sqft
Prestige Group 8400000 7200
Brigade Group 17500000 7500
Brigade Group 21000000 12000
Total 46900000
sales trend
20M
10M
total revenue
15 Jan
05 Mar
Sale_Date
"Demand vs Supply by Area"
Count of Demand_Index
Count of Supply_Index
Count of Demand_I...
Electronic City
Indiranagar
Sarjapur
Whitefield
Area
Count of Property_ID by Property_Type
1 (20%)
1 (2...)
3 (60%)
Property_Type
Apartme...
Plot
Villa

CONCLUSION

The *Real Estate Data Analysis Dashboard* effectively demonstrates how property-related data can be analyzed and visualized to generate meaningful insights. By using data on properties, builders, customers, and transactions, the project provides a clear understanding of overall market performance, pricing trends, and builder contributions across different areas.

The integration of interactive charts, KPI cards, and dynamic filters allows users to explore the data from multiple perspectives. Users can identify high-value areas, compare builder performance, and observe trends in property sales, making the analysis more informative and practical.

The project also highlights the importance of data cleaning, transformation, and visualization in building effective analytical dashboards. It demonstrates practical knowledge of Power BI tools such as Power Query for data preparation, DAX for calculations, and dashboard design techniques for creating clear and interactive visuals.

In conclusion, the project successfully fulfills its objective of developing an intuitive and informative dashboard for real estate data analysis. It serves as a useful tool for understanding market trends, supporting decision-making, and learning business intelligence concepts. The project can be further enhanced by incorporating real-time data, predictive analysis, and more detailed geographic insights in the future.

REFERENCES

[1] Real Estate Dataset, "Property Transactions and Pricing Data," 2026. [Dataset].

[2] Microsoft Corporation, "Power BI Documentation," Microsoft Learn, 2026. [Online]. Available: https://learn.microsoft.com/power-bi/. [Accessed: May 2, 2026].

[3] Microsoft Corporation, "DAX Function Reference," Microsoft Learn, 2026. [Online]. Available: https://learn.microsoft.com/dax/. [Accessed: May 2, 2026].

[4] Alberto Ferrari and Marco Russo, *The Definitive Guide to DAX*, 2nd ed. Redmond, WA, USA: Microsoft Press, 2020.

[5] Microsoft Corporation, "Power Query Documentation," Microsoft Learn, 2026. [Online]. Available: https://learn.microsoft.com/power-query/. [Accessed: May 2, 2026].

[6] Cole Nussbaumer Knaflic, *Storytelling with Data: A Data Visualization Guide for Business Professionals*. Hoboken, NJ, USA: Wiley, 2015.

[7] Online Tutorials and Learning Resources, "Power BI Dashboard Development," 2026.